Dashboard `DeviceStatus.health`-is-an-object regression guard — QA evidence

Run-id: `20260610T113752Z-dashboard-health-guard` **Date (UTC):** 2026-06-10T11:38:28Z **Scope:** dashboard/ Vitest component layer **Constitution:** §11.4.135 (every fixed defect gets a permanent regression guard) · §11.4.115 (RED-on-broken-shape → GREEN-on-fixed-shape with a `RED_MODE` polarity switch) · §11.4 anti-bluff (real Vitest runs captured in `vitest_run.txt`).

The product bug (closed by commit 4658125, shipped WITHOUT a guard)

`commit 4658125 — "fix(dashboard): DeviceStatus.health is an object — fix Fleet-detail crash + cover"`

The server wire shape of `DeviceStatus.health` is an **OBJECT** `{ ok: bool, last_error_code: string|null }` (server `internal/api/wire.go` `DeviceHealth`). The dashboard wrongly typed it `health?: string` and the Fleet device-detail screen (`FleetScreen.tsx` `DeviceDetail`) rendered it as a bare React child:

```
<Row label="health" value={status.data.health ?? "—"} />
```

With the real object shape React throws **"Objects are not valid as a React child (found: object with keys {ok, last_error_code})"**, the `<DeviceDetail>` render throws, and the **entire device page blanks** for the operator (a §11.4.108 SOURCE→USER-VISIBLE failure). The fix added the `DeviceHealth` type and projects `health.ok` → an ok/unhealthy badge + `health.last_error_code` text.

The commit shipped tests but no §11.4.135 RED/GREEN polarity guard mechanically distinguishing the pre-fix from post-fix render path. This work closes that governance gap.

The guard

`dashboard/src/screens/FleetScreen.health-regression.test.tsx` — one `RED_MODE` polarity switch (default `0`):

- **RED_MODE=1** — reproduces the PRE-FIX render path verbatim (`<dd>{health as React.ReactNode}</dd>` with the *real* object-shaped `health`) and asserts React **does throw** `/Objects are not valid as a`

`React child/`. This is the §11.4.115 RED-on-the-broken-shape proof that the defect is real and the guard genuinely catches it.
- **RED_MODE=0** (default — the standing GREEN guard) — renders the CURRENT shipped `<DeviceDetail>` with the real object-shaped `health` and asserts it (a) does NOT crash, (b) projects the object to text ("ok" / "unhealthy" + `last_error_code`), (c) never leaks `"[object Object]"` into the DOM, plus a negative-shape (`ok:false`) case and a compile-time type-contract lock.

If a future regression re-types `health` to a scalar or re-renders the object directly, the GREEN assertions FAIL.

Commands run + results (full transcript: `vitest_run.txt`)

Command	Result
`npx vitest run src/screens/FleetScreen.health-regression.test.tsx` (GREEN default)	**3 passed**
`RED_MODE=1 npx vitest run src/screens/FleetScreen.health-regression.test.tsx`	**1 passed** — the throw was asserted; pre-fix crash reproduced
`npx vitest run` (full suite)	**58 passed** (was 55 at commit 4658125; +3 new guards)
`npm run typecheck`	**clean** (tsc app + node + vitest tsconfigs)

Tooling: node v22.22.3, npm 10.9.8, Vitest 2.1.9, node_modules already installed from the committed package-lock.json.